

Problem Set 07

Your Name

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Background

For this exercise, we will mimic the tactile sampling you did in class with virtual sampling. We will use some data from the [general social survey](#), an annual personal-interview survey conducted in the United States. The survey is designed to monitor changes in both social characteristics and attitudes.

The **population** of interest will be **ALL** 2538 individuals living in a single neighborhood in 2014. As an analogy to the tactile sampling you did in class, the neighborhood is the “bowl” and the 2538 people are the little balls.

If you get stuck as you are working through this Problem Set, we **strongly recommend** you re-read [Chapter 7](#) in [ModernDive](#), in particular subsections 7.3.1 on “[Terminology & notation](#)” and 7.3.2 on “[Statistical definitions](#)”. These terminology, notation, and definitions related to sampling are definitely tricky at first; the best method to master them is practice, practice, practice.

Key points on symbols:

Symbol	POPULATION PARAMETER	SAMPLE STATISTIC
Number of cases	N	n
Proportion	p	$\hat{p}$
Standard error	SE	$\widehat{SE}$

R Code

```
sym <- c("Number of cases", "Proportion", "Standard error")
PP <- c("$N$", "$p$", "$SE$")
SS <- c("$n$", "$\\hat{p}$", "$\\widehat{SE}$")
DF <- data.frame(sym = sym, PP = PP, SS = SS)
kable(DF, col.names = c("Symbol", "Population Parameter",
                        "Sample Statistic"),
      caption = "Key Points on Symbols")
```

Table 2: Key Points on Symbols

Symbol	Population Parameter	Sample Statistic
Number of cases	N	n
Proportion	p	$\hat{p}$
Standard error	SE	$\widehat{SE}$

Setup

First load the necessary packages:

R Code

```
library(ggplot2)
library(dplyr)
library(forcats)
library(moderndiver)
```

You can load and view the `gss_cat` data set from the `forcats` package using the code below:

R Code

```
data(gss_cat)
glimpse(gss_cat)
```

Rows: 21,483

Columns: 9

```
$ year      <int> 2000, 2000, 2000, 2000, 2000, 2000, 2000, 2000, 2000, 2000, 20~
$ marital   <fct> Never married, Divorced, Widowed, Never married, Divorced, Mar~
$ age       <int> 26, 48, 67, 39, 25, 25, 36, 44, 44, 47, 53, 52, 52, 51, 52, 40~
$ race      <fct> White, White, White, White, White, White, White, White, White, ~
$ rincome   <fct> $8000 to 9999, $8000 to 9999, Not applicable, Not applicable, ~
$ partyid   <fct> "Ind,near rep", "Not str republican", "Independent", "Ind,near~
$ relig     <fct> Protestant, Protestant, Protestant, Orthodox-christian, None, ~
$ denom     <fct> "Southern baptist", "Baptist-dk which", "No denomination", "No~
$ tvhours   <int> 12, NA, 2, 4, 1, NA, 3, NA, 0, 3, 2, NA, 1, NA, 1, 7, NA, 3, 3~
```

Be sure to examine the data in the **viewer**. Type `?gss_cat` in the **console** to see a description of the variables in this data set.

Exploratory Data Wrangling

This data set includes many of years of data and many variables. To start, we will restrict our analysis to only 2014, and to only the variable indicating the `marital` status of each respondent.

R Code

```
YEAR

[1] 2014

gss_YEAR <- gss_cat |>
  filter(year == YEAR) |>
  select(marital)
```

The following shows the different responses for `marital` status:

R Code

```
gss_YEAR |>
  distinct(marital)

# A tibble: 6 x 1
  marital
  <fct>
```

```
1 Divorced
2 Married
3 Never married
4 Separated
5 Widowed
6 No answer
```

Setting a Seed

Setting a seed: We will take some random samples in this Problem Set. In order to make sure R takes the same random sample every time you run your code, you can do what is called “setting a seed”. Do this in any code chunk where you take a random sample.

You can set a seed with any number.

```
set.seed(45)
```

The True Population Proportion p of Divorced People

Again, for this exercise, the **population** of interest will be **ALL** 2538 individuals living in this single neighborhood in 2014. Since we have data on **ALL** 2538 people living in the neighborhood, we can compute the **exact population proportion** p of divorced people directly using **ALL** the data as follows:

R Code

```
gss_YEAR |>
  summarize(divorced = sum(marital == "Divorced"),
            N = n(), p2 = mean(marital == "Divorced")) |>
  mutate(p = divorced / N) -> ans
ans

# A tibble: 1 x 4
  divorced      N    p2      p
  <int> <int> <dbl> <dbl>
1     411  2538 0.162 0.162

# Or
(p <- mean(gss_YEAR$marital=="Divorced"))
```

```
[1] 0.1619385
```

i Note

- We use N for the size of the full population of 2538 people, and p because we are calculating the TRUE population proportion p .
- Inference to the population is not needed. We do not need to use a **sample** to try to infer something about the **true population proportion** p of divorced people in this neighborhood in 2014. We know that $p = 0.1619385$.

In other words, this problem set is not a realistic reflection of a real life problem. However, for this problem set, we are *simulating* the act of sampling from this neighborhood population to understand and study how factors like sample size influence **sampling variation**.

Demo: Sampling 50 People in the Neighborhood

Estimating $\hat{p}$ from a Single Sample

We are first going to use random sampling to **ESTIMATE** the true **population** proportion p of the neighborhood that are divorced with only a **sample** of 50 people.

i Note

This will represent a situation of only having the resources to knock on 50 doors to get responses from people in this neighborhood!

Be sure to look at the results in the viewer. Remember, you can set the seed to whatever value you like. For this exercise, use a four digit seed that is your birth month and day. For example, if your birthday is on Halloween then you will use the seed 1031.

R Code

```
set.seed(0804)
n50_1rep <- gss_YEAR |>
  rep_sample_n(size = 50, reps = 1)
head(n50_1rep)

# A tibble: 6 x 2
# Groups:   replicate [1]
```

```

replicate marital
  <int> <fct>
1      1 Never married
2      1 Never married
3      1 Married
4      1 Never married
5      1 Never married
6      1 Married

# OR
set.seed(0804)
n50_sample <- sample(gss_YEAR$marital, size = 50, replace = FALSE)
head(n50_sample)

[1] Never married Never married Married      Never married Never married
[6] Married
Levels: No answer Never married Separated Divorced Widowed Married

```

Next, let's calculate the **sample proportion** $\hat{p}$ of people who identified as Divorced in our sample of 50 people.

R Code

```

ans1 <- n50_1rep |>
  summarize(divorce_count = sum(marital == "Divorced"),
            phat = mean(marital == "Divorced"),
            n = n()) |>
  mutate(p_hat = divorce_count/ n)
ans1

# A tibble: 1 x 5
  replicate divorce_count phat      n p_hat
  <int>      <int> <dbl> <int> <dbl>
1         1         6 0.12    50 0.12

# OR
(phat <- mean(n50_sample=="Divorced"))

[1] 0.12

```

This sample proportion $\hat{p}$ is an **ESTIMATE**, and our **best guess** of what the **true popu-**

lation proportion p of **Divorced** people is in this neighborhood, based on a sample of only 50 people. The estimate ($\hat{p} = 0.12$) is reasonably close to the true population proportion $p = 0.1619385$ we calculated from the full population.

Problem 1

Ask two of your classmates what their estimate of $\hat{p}$ was. How do the $\hat{p}$ estimates from different samples compare?

Problem 1 Answers

- Delete this and put your text answer here.

Problem 2

Why did everyone get a different estimate?

Problem 2 Answers

- Delete this and put your text answer here.

Estimating $\widehat{SE}$ from a Single Sample

Typically we only have the opportunity to collect **one sample** for our study. Consequently, we have to use the amount of variability in our **single sample** as an estimate of the amount of variability we might expect in our results if we had taken a random sample of 50 different people. The $\widehat{SE}_{\hat{p}}$ serves as an **ESTIMATE** of **sampling variability** if you only have a **single sample**. The formula for estimating the standard error of $\hat{p}$ is given in (Equation 1).

$$\widehat{SE}_{\hat{p}} \approx \sqrt{\frac{\hat{p} \times (1 - \hat{p})}{n}} \quad (1)$$

i Note

Note that we use n for the size of the sample, that p “wears a hat”, like so: $\hat{p}$ because we are **ESTIMATING** a proportion based on only a sample, and that the SE “wears a hat” as well because we are **ESTIMATING** SE based on only a sample.

The standard error of $\hat{p}$ can be estimated in R as follows:

R Code

```
n50_1rep |>
  summarize(divorce_count = sum(marital == "Divorced"),
            n = n()) |>
  mutate(p_hat = divorce_count / n,
         se_hat = sqrt(p_hat * (1 - p_hat) / n)) -> n50_1rep_ans

# A tibble: 1 x 5
  replicate divorce_count      n p_hat se_hat
  <int>      <int> <int> <dbl> <dbl>
1         1          6    50  0.12 0.0460
```

Demo: Generating a Sampling Distribution of $\hat{p}$

If you ran the code chunk that takes a random sample of 50 cases a thousand more times....and wrote down every $\hat{p}$ you got, you would have what is called a simulated “sampling distribution”.

Note

A sampling distribution shows every [or nearly every!] possible result a sampling statistic can have under every [or nearly every!] possible sample **of a given sample size** from a population.

Simulated Sampling Distribution of $\hat{p}$ for $n = 50$

Instead of running the sampling code chunk for $n = 50$ over and over, we can “collect” 1000 samples of $n = 50$ easily with R. The following code chunk takes 1000 **different** samples of $n = 50$ and stores them in the data frame `n50_1000rep`:

R Code

```
set.seed(19)
n50_1000rep <- gss_YEAR |>
  rep_sample_n(size = 50, reps = 1000)
head(n50_1000rep)
```

```

# A tibble: 6 x 2
# Groups:   replicate [1]
  replicate marital
      <int> <fct>
1         1 Divorced
2         1 Never married
3         1 Never married
4         1 Widowed
5         1 Married
6         1 Widowed

# Consider a different approach
set.seed(19)
reps <- 1000
n <- 50
phat <- numeric(reps)
for(i in 1:reps){
  ss <- sample(gss_YEAR$marital, size = 50, replace = FALSE)
  phat[i] <- mean(ss == "Divorced")
}
head(phat)

[1] 0.08 0.10 0.16 0.18 0.20 0.10

```

Be sure to look at `n50_1000rep` in the data viewer to get a sense of these 1000 samples look like.

Problem 3

What is the name of the column that identifies which of the 1000 samples each row is from?

Problem 3 Answers

- Delete this and put your text answer here.

Problem 4

What is the sample size n for each of the 1000 samples we took? (i.e. how many humans are sampled in each replicate)?

Problem 4 Answers

- Delete this and put your text answer here.

The following code chunk calculates the sample proportion $\hat{p}$ of people who reported they were divorced for each of the **1000 samples**.

R Code

```
p_hat_n50_1000rep <- n50_1000rep |>
  group_by(replicate) |>
  summarize(divorce_count = sum(marital == "Divorced"),
            n = n()) |>
  mutate(p_hat = divorce_count / n)
```

Examine the first five rows of the results:

R Code

```
p_hat_n50_1000rep |>
  slice(1:5)
```

```
# A tibble: 5 x 4
  replicate divorce_count      n p_hat
  <int>         <int> <int> <dbl>
1         1           4    50  0.08
2         2           5    50  0.1
3         3           8    50  0.16
4         4           9    50  0.18
5         5          10    50  0.2
```

```
# Or
p_hat_n50_1000rep |>
  head(n = 5)
```

```
# A tibble: 5 x 4
  replicate divorce_count      n p_hat
  <int>         <int> <int> <dbl>
1         1           4    50  0.08
2         2           5    50  0.1
3         3           8    50  0.16
```

4	4	9	50	0.18
5	5	10	50	0.2

Visualize the Sampling Distribution of $\hat{p}$ for $n = 50$

We can plot the **sampling distribution** of these 1000 $\hat{p}$ estimates of divorced respondents with a histogram using the code below.

R Code

```
# histogram of phat using ggplot2
ggplot(p_hat_n50_1000rep, aes(x = p_hat)) +
  geom_histogram(binwidth = 0.02, color = "black", fill = "aquamarine3") +
  labs(x = "Sample proportion of divorced respondents",
       title = "Sampling distribution of p_hat based on n = 50") +
  theme_bw() +
  theme(plot.title = element_text(hjust = 0.5)) +
  xlim(-0.05, 0.6) -> hist1
hist1
```

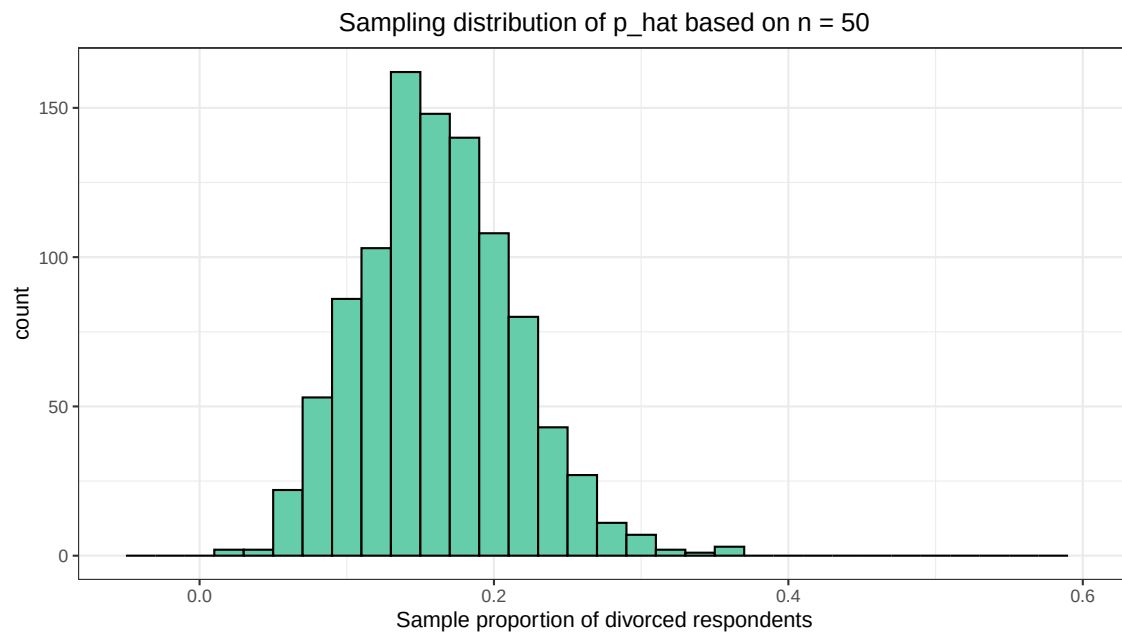


Figure 1: Simulated sampling distribution of $\hat{p}$ based on 1000 samples of size 50 created with ggplot2

```
# hist of phat using base R
hist(phat, main = "Sampling distribution of phat based on n = 50",
     col = "aquamarine3", breaks = 15)
```

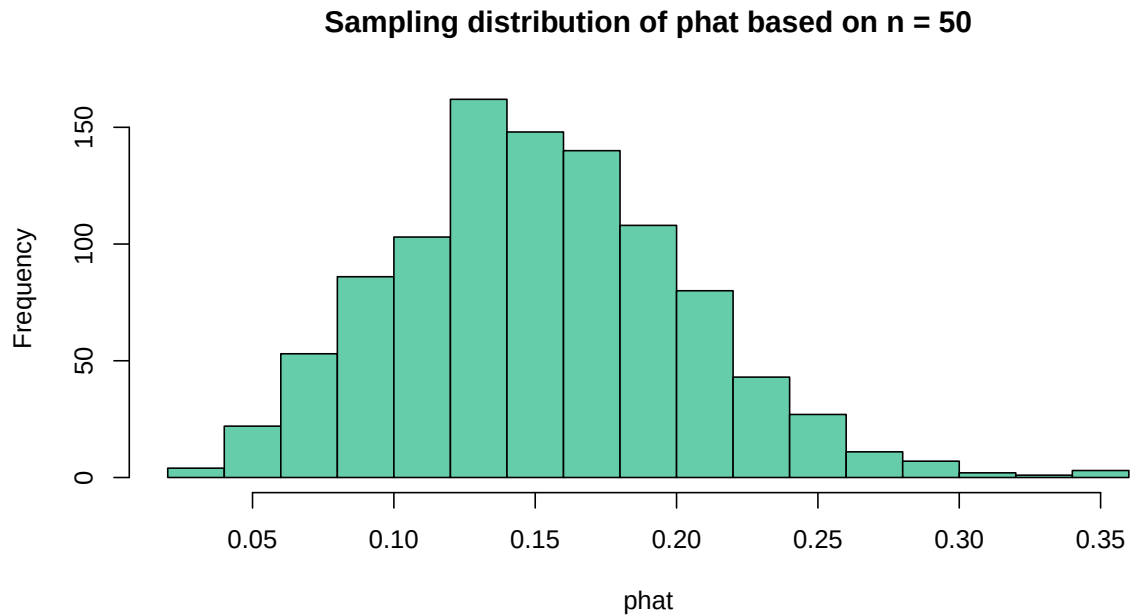


Figure 2: Simulated sampling distribution of $\hat{p}$ based on 1000 samples of size 50 created with base R

Problem 5

Based on Figure 1 and Figure 2, what appears to be a very common value of $\hat{p}$? What is a very uncommon value? Specifically, find the 1%, 99%, the mean, and the standard deviation of the values stored in `p_hat` to help answer the question.

Problem 5 Answers

```
# Type your code and comments below
```

- Delete this and put your text answer here.

Mean and Standard Error of the Sampling Distribution of $\hat{p}$ for $n = 50$

We can estimate the mean of the sampling distribution by calculating the mean of all 1000 $\hat{p}$ estimates, and the standard error of the sampling distribution by calculating the standard deviation of all 1000 $\hat{p}$ values as follows:

R Code

```
p_hat_n50_1000rep |>
  summarize(M_p_hat = mean(p_hat),
            SE_p_hat = sd(p_hat))

# A tibble: 1 x 2
  M_p_hat SE_p_hat
  <dbl>   <dbl>
1  0.162   0.0522
```

Problem 6

How do the values computed in the above code chunk compare to the estimates we got for $\hat{p}$ and $\widehat{SE}_{\hat{p}}$ for **Divorced** respondents based on your **single** sample of 50 people earlier in this Problem Set?

Problem 6 Answers

- Delete this and put your text answer here.

Problem 7

Use the `rep_sample_n` function to collect 1000 virtual samples of size $n = 15$. Store the 1000 virtual samples in an object named `n15_1000rep`. Use a seed of 910.

Problem 7 Answers

```
# Type your code and comments inside the code chunk
```

Problem 8

Calculate sample proportion $\hat{p}$ of people who reported they were **Divorced** for each replicate of your $n = 15$ sampling. Store the results in `ques8` and display the first six

rows of ques8.

Problem 8 Answers

```
# Type your code and comments inside the code chunk
```

Problem 9

Visualize the sampling distribution of $\hat{p}$ from your $n = 15$ sampling with a purple histogram.

Problem 9 Answers

```
# Type your code and comments inside the code chunk
```

Problem 10

Calculate the mean of the $n = 15$ sampling distribution, and the standard error of the $n = 15$ sampling distribution.

Problem 10 Answers

```
# Type your code and comments inside the code chunk
```

Problem 11

How does the standard error of the $n = 15$ sampling distribution compare to the standard error of the $n = 50$ sampling distribution?

Problem 11 Answers

- Delete this and put your text answer here.

Problem 12

Explain any observed differences from 11.

Problem 12 Answers

- Delete this and put your text answer here.

Problem 13

Use the `rep_sample_n` function to collect 1000 virtual samples of size $n = 600$. Store the 1000 virtual samples in an object named `n600_1000rep`. Use a seed of 84.

Problem 13 Answers

```
# Type your code and comments inside the code chunk
```

Problem 14

Calculate the proportion $\hat{p}$ of people who reported they were `Divorced` for each replicate of your $n = 600$ sampling. Store the results in `ques14` and display the first six rows of `ques14`.

Problem 14 Answers

```
# Type your code and comments inside the code chunk
```

Problem 15

Calculate the mean of the $n = 600$ sampling distribution, and the standard error of the $n = 600$ sampling distribution.

Problem 15 Answers

```
# Type your code and answers below
```

Problem 16

Was there more **variability** from sample to sample when we took a sample size of 600 or a sample size of 50? **Explain what evidence you have for assessing this.**

Problem 16 Answers

- Delete this and put your text answer here.

Problem 17

Which sampling distribution looked more normally distributed (bell shaped and symmetrical); the one built on $n = 15$, 50 or 600? **Why?**

Problem 17 Answers

- Delete this and put your text answer here.

```
# Type your code and comments below
```

Estimating $\hat{p}$ and the Standard Error of $\hat{p}$ from a Single Sample (revisited)

In most instances, we do not have access to the full population as we did in this GSS data; instead we have to take a **sample** to try to say something about the **larger population**. Furthermore, in the real world, we typically only take a **single** sample from the population, due to time or money constraints.

So how do we **ESTIMATE** a $\hat{p}$ and a standard error of $\hat{p}$ when we only have a single sample, and not 1000 repeated samples? As demonstrated at the very beginning of the Problem Set we:

- estimate $\hat{p}$ from the sample
- use the formula for the standard error of $\hat{p}$ given in (Equation 1) and repeated below to estimate SE based on a single sample

$$\widehat{SE}_{\hat{p}} \approx \sqrt{\frac{\hat{p} \times (1 - \hat{p})}{n}}$$

Problem 18

Imagine we collected only a single small sample of 15 respondents as given from the code below.

```
set.seed(53)
n15_1rep <- gss_YEAR |>
  rep_sample_n(size = 15, reps = 1)
# and
n50_1rep <- gss_YEAR |>
  rep_sample_n(size = 50, reps = 1)
```

Following the example from the beginning of the Problem Set (roughly line 138), estimate the **sample proportion** $\hat{p}$ of people who identified as Divorced based on `n15_1rep...` AS WELL AS the **standard error of $\hat{p}$** .

Problem 18 Answers

```
# Type your code and comments inside the code chunk
```

Note

You should have gotten a value reasonably close to the estimate we made earlier from our sampling distribution for $n = 15$! Note that when you must estimate a standard error from **only a single sample**, the formula **contains the sample size, n** . The larger the sample size n , the larger the number in the denominator of the SE formula.

Fill in the Markdown table below with all the standard errors you computed for this problem set. In other words:

Problem 19

- Replace **x** with the standard error you obtained by taking the standard deviation of the $n = 15$ sampling distribution. Replace **a** with the standard error you obtained for a single sample of $n = 15$ using the mathematical formula.
- When you are done, make sure all the | in the table still line up so your results print out in a table!

Problem 19 Answers

Sample size n	SE via sd of sampling distribution	SE via one sample and formula
15	x	a

50

y

b

```
# Your code below
```

Problem 20

Based on what you observed for 19, **IF** you collected a single sample from 600 respondents, do you think the standard error will be smaller or larger than the one you calculated for $n = 15$. **Explain your reasoning.**

Problem 20 Answers

- Delete this and put your text answer here.

Turning in Your Work

You will need to make sure you commit and push all of your changes to the github education repository where you obtained the lab.

Tip

- Make sure you **render a final copy with all your changes** and work.
- Look at your final html file to make sure it contains the work you expect and is formatted properly.

Logging out of the Server

There are many statistics classes and students using the Server. To keep the server running as fast as possible, it is best to sign out when you are done. To do so, follow all the same steps for closing Quarto document:

Tip

- Save all your work.
- Click on the orange button in the far right corner of the screen to quit R
- Choose **don't save** for the **Workspace image**
- When the browser refreshes, you can click on the sign out next to your name in the

- top right.
- You are signed out.

```
sessionInfo()
```

```
R version 4.4.1 (2024-06-14)
Platform: x86_64-redhat-linux-gnu
Running under: Red Hat Enterprise Linux 9.4 (Plow)

Matrix products: default
BLAS/LAPACK: FlexiBLAS OPENBLAS-OPENMP; LAPACK version 3.9.0

locale:
 [1] LC_CTYPE=en_US.UTF-8      LC_NUMERIC=C
 [3] LC_TIME=en_US.UTF-8      LC_COLLATE=en_US.UTF-8
 [5] LC_MONETARY=en_US.UTF-8  LC_MESSAGES=en_US.UTF-8
 [7] LC_PAPER=en_US.UTF-8     LC_NAME=C
 [9] LC_ADDRESS=C             LC_TELEPHONE=C
[11] LC_MEASUREMENT=en_US.UTF-8 LC_IDENTIFICATION=C

time zone: America/New_York
tzcode source: system (glibc)

attached base packages:
[1] stats      graphics  grDevices  utils      datasets  methods   base

other attached packages:
[1] moderndive_0.6.1 forcats_1.0.0    dplyr_1.1.4      ggplot2_3.5.1
[5] knitr_1.48

loaded via a namespace (and not attached):
 [1] gtable_0.3.5      jsonlite_1.8.8    compiler_4.4.1
 [4] tinytex_0.52      tidyrselect_1.2.1 stringr_1.5.1
 [7] snakecase_0.11.1  tidyr_1.3.1       scales_1.3.0
[10] yaml_2.3.10       fastmap_1.2.0     R6_2.5.1
[13] labeling_0.4.3    generics_0.1.3    infer_1.0.7
[16] operator.tools_1.6.3 backports_1.5.0    tibble_3.2.1
[19] janitor_2.2.0     munsell_0.5.1     lubridate_1.9.3
[22] pillar_1.9.0      formula.tools_1.7.1 rlang_1.1.4
[25] utf8_1.2.4        stringi_1.8.4     broom_1.0.6
[28] xfun_0.47         timechange_0.3.0  cli_3.6.3
```

[31]	withr_3.0.1	magrittr_2.0.3	digest_0.6.36
[34]	grid_4.4.1	rstudioapi_0.16.0	lifecycle_1.0.4
[37]	vctrs_0.6.5	evaluate_0.24.0	glue_1.7.0
[40]	farver_2.1.2	fansi_1.0.6	colorspace_2.1-1
[43]	rmarkdown_2.28	purrr_1.0.2	tools_4.4.1
[46]	pkgconfig_2.0.3	htmltools_0.5.8.1	